

Plan of work — rev 2

O&M Cane Training · data & upload architecture · reconciled with external code review

This revision corrects rev 1, which described the data-key work as ‘settled’ when it was decided but **not yet implemented**. An external developer review of the live source confirmed the gap. The ‘settled’ framing has been split into **decided** vs. **in code** below.

CORRECTION FROM REV 1

Rev 1 listed ‘stamp childId + teacherId on every record from v1’ as **settled**. In the actual code: **teacherId does not exist**, and childId is **profileId** — a device-local random ID, not stable across devices. The decision was made; the implementation was not. Rev 1 read as more done than git reality. This document now tracks **decided** and **in code** as separate columns so no reviewer is misled again.

ARCHITECTURE DECISION (unchanged, confirmed sound)

Option B — reviewable session archive: sessions upload to India-region Supabase with their context (activity, result, notes) for the R&D team (NCAHT / IIT Delhi) to review and improve the assessment. Identified video (faces visible) is confirmed required by R&D. The review endorsed B and the ‘identified video sets the handling bar regardless of shape’ reasoning. Final A-vs-B confirmation from R&D still pending, but B is the working assumption.

DECIDED vs. IN CODE

Item	Decided?	In code?	Un-backfillable?
teacherId on every record	Yes	NO — absent entirely	YES — must precede any production session
Stable, device-stable childId	Yes	Partial — profileId is device-local only	YES
Stable per-record id (UUID)	Yes	NO — delete is index-based	YES
ISO 8601 timestamp (whenISO)	Yes	NO — only locale string	YES
schemaVersion on records	Yes	NO	YES (cheap, eases B→A)
Operator device lock (PIN)	Yes	NO	No — addable later, but precedes identified video
Delete-on-withdrawal (local)	Yes	Partial — 3-tier delete exists locally	n/a
Cloud uploader + duties	Yes	NO — not built	n/a

WHAT THE REVIEW CONFIRMED AS STRONG

The **Store seam** (sync-cache over async backend, write-through + read-back verify) was called the standout — the right template for the planned Uploader seam. **Deletion** is real and three-tier locally. **CSV export** is robust (RFC-4180 quoting, UTF-8 BOM, native Filesystem→Share path). **XSS discipline** via `esc()` is consistently applied. These are not at risk and the Uploader should copy the Store seam's verify discipline.

BUILD ORDER (revised by trigger point, not lumped)

The review's key sequencing insight: separate the **un-backfillable** items (corrupt or impossible if real data exists first) from the merely-important. Three trigger points:

Batch 1 — before ANY production session (un-backfillable):

- Mint + stamp a per-install **teacherId** at boot (behind the Store seam).
- Give every record a stable **UUID id**; switch deletion from delete-by-index to **delete-by-id**.
- Store **whenISO** (ISO 8601) as the canonical timestamp; derive the display string at render.
- Add **schemaVersion** to records/profiles.

Batch 2 — correctness fixes (soon, before identified data):

- Replace **window.confirm()** with an in-DOM focus-trapped modal (type-to-confirm for full wipe).
- Harden **boot failure**: set `Store._failed`, hard-block writes, show a blocking 'storage unavailable' state instead of silently succeeding against an empty cache.
- Rename the photo 'limits identifiability' comment to 'limits storage size' — a 160px face is still identifiable; don't let it mislead the DPIA.

Batch 3 — before identified video flows:

- Operator **device lock** (PIN on launch + re-gate before danger zone / identified export).
- Local **photo-at-rest** handling decided (encrypt, or drop pre-upload in favour of server-side media).

Batch 4 — before the Uploader is built (cloud architecture, locked):

- **Private** Supabase bucket; access only via short-TTL **signed URLs** minted server-side after an authz check — never a public object URL.
- Metadata in Postgres with **Row-Level Security** keyed on `teacherId` / institution; video in Storage, joined by key.
- **Cascade delete** (DB row + storage object + thumbnails) implemented and **proven by a test** — orphaned objects are a DPDP finding.
- Pin **India region** explicitly; confirm no cross-region replication.
- **Queue-and-retry** upload (intermittent classroom connectivity), with the same write-verify discipline as the Store seam.

Engineering plan, not legal advice. Consent text + the app-to-institution data-sharing agreement need qualified review before any real child data is collected. App handles children's data under India's DPDP Act 2023 + Rules 2025.